Oceans and Climate

EAS 4850/6850 Ocean Circulation and Climate

Spring Semester 2027

Instructor: Prof. Una Miller, ukm6@cornell.edu

Credits: 3

Class Schedule: Tu/Thr 1:25pm – 2:40pm

Location: TBD

Office hours: Email Prof. Miller to schedule

Course sections: EAS 4850 (undergraduate) and EAS 6850 (graduate) meet together. Graduate students complete additional requirements, described under "EAS 6850 (graduate section) requirements" below

Prerequisites: ONE semester of calculus (MATH 1110 or MATH 1910, or equivalent) and ONE semester of introductory physics (PHYS 1101, PHYS 1112, or PHYS 2207, or equivalent), or permission of the instructor. No prior coursework in oceanography is required.

Textbook: Climate and the Oceans by Geoffrey Vallis. Available as an e-book through CAMP/Instant Access.

Course description: We will build a descriptive understanding of the dynamics that govern ocean circulation in order to explore the ocean's role in Earth's climate system. The course emphasizes large-scale ocean circulation, with a particular focus on the Atlantic Meridional Overturning Circulation (AMOC) and Southern Ocean Meridional Overturning Circulation (SOMOC) and the role of polar and subpolar processes in these systems. We will engage with contemporary scientific literature to examine how ocean circulation is changing and how these changes impact the redistribution of heat, carbon, and oxygen within the climate system. Intended for students in Earth and Climate Sciences, Atmospheric Science, Engineering, Marine Biology, and other science majors. Prior coursework in oceanography is not assumed.

Learning outcomes: As a result of participating in this class, students will be able to:

- Describe the role of large-scale ocean circulation in global heat and carbon redistribution, and its impact on climate

- Explain observed and projected changes in large-scale ocean circulation in a warming climate, including points of consensus and debate within the scientific community
- Critically interpret the motivations, methods, and key findings of scientific journal papers
- Communicate scientific knowledge effectively, both orally and in writing

Grading:

15% Canvas reading quizzes

20% In-class paper discussions

10% In-class activities

20% In-class quizzes

15% Final presentation

20% Final Exam

Final grades are computed from percent of total possible points. Letter grades are assigned as follows:

Grade	Percentage	Grade	Percentage
A+	97–100	C+	77–79.99
A	93–96.99	C	73–76.99
A−	90–92.99	C−	70–72.99
B+	87–89.99	D+	67–69.99
B	83–86.99	D	63–66.99
B−	80–82.99	D−	60–62.99
		F	Below 60

Assignments (EAS 4850; undergraduate section):

Canvas reading quizzes: Quizzes on Canvas are based on the assigned readings and are intended to help you prepare for lectures and in-class discussions. They are due at class time the day of the discussion. *Late submission of Canvas quizzes will not be accepted, however, your lowest quiz score will be dropped from your final grade.*

In-class paper discussion: Worksheets will be handed out on the days we have paper discussions, and worked on together in small groups to be turned in by the end of class. Graduate students in EAS 6850 will lead the small discussion groups on these days (see the EAS 6850 section below). *If you must miss an in-class paper discussion day, please notify Prof. Miller by email. You will be responsible for completing and submitting the in-class worksheet within a week.*

In-class activities: On most class days, we will have a short 1- or 2-question activity based on lecture material. Answers are written on index cards, which must be turned in at the end of class. These are graded on participation only. *If you miss class, you will receive a 0 for the activity, however, your lowest two scores will be dropped from your final grade.*

In-class quizzes: There will be three quizzes, covering all prior lecture material. *The lowest of your three quiz scores will be dropped from your final grade, however, this does NOT mean you should skip a quiz because you feel unprepared or have something you'd rather do on quiz day. If you must miss a quiz, email Prof. Miller as soon as possible with your valid reason for your absence. Please see Late and Missed Assignments section for valid reasons.*

Final presentation: Students will select a scientific paper relevant to material covered in the course and present its motivations, methods, and key findings, as well as their critical assessment of the paper's strengths and limitations to the class. Presentations should include an engaging question or activity for the audience similar to the style of in-class activity questions asked by Prof. Miller throughout the course. Undergraduates present in pairs; graduate students present individually. Graduate students will additionally place their chosen paper in the context of 2-3 related papers and identify an important unresolved question or direction for future research. Undergraduate presentations will be 10-15 minutes with 5 minutes for questions and graduate presentations will be 20-25 minutes, with 5 for questions. Presentations will be evaluated on how well the students:

Criterion	**EAS 4850**	**EAS 6850**
Explain the motivation for the study — the scientific question the paper addresses and why it matters	20%	10%
Explain the methods clearly and accurately, including the data, models, or analyses used and why those choices suit the question	15%	10%
Present the key findings accurately, including a clear walk-through of at least one main figure	20%	15%

Criterion	EAS 4850	EAS 6850
Offer a critical assessment of the paper: its strengths, limitations, and underlying assumptions	20%	15%
Deliver an organized, clearly designed presentation within the allotted time	10%	10%
Respond thoughtfully to questions from the audience	5%	5%
Engage the audience with a question during the presentation	10%	10%
Use 2–3 additional papers to provide context	—	15%
Present a direction for future research	—	10%
Total	**100%**	**100%**

Final exam**:** The final exam will be a handwritten, in-person essay exam held during finals week at the day and time assigned by the University Registrar. It will ask a synthesis question based on the readings and discussions from the course. Students will be expected to explicitly reference the four papers assigned during the semester in their essay and should come prepared with printouts of the four papers. Grading will be based on scientific accuracy (25%), the use of specific evidence to support claims (25%), the quality of reasoning in weighing that evidence (25%), and completeness in addressing all parts of the prompt (25%). Length is not graded. If printing cost is a concern, please speak with Prof. Miller. Phones and laptops will not be allowed.

Assignments (EAS 6850; graduate section): Graduate students attend the same class sessions, complete the same readings, reading quizzes, worksheets, in-class activities, and quizzes as undergraduates, and are held to the same grading scale. They will be assigned additional work in the following categories:

In-class paper discussion – 6850 additional requirements:

Leading discussions: On paper discussion days, graduate students lead the small discussion groups: they come prepared with the paper read closely, guide their group through the worksheet questions, and help the group articulate what the paper does and does not establish. In addition, each graduate student submits two or three discussion questions of their own on Canvas by the start of class. These should be open-ended questions that invite interpretation or debate — about the strength of the evidence, the authors' assumptions, or how the paper fits with others we have read — rather than questions with a single factual answer. Graduate students pose their questions to their group once the worksheet is underway, and Prof. Miller

may bring some of them to the full class. Prof. Miller will meet briefly with graduate students before the first discussion day to go over expectations.

Complementary papers: For each paper discussion day, graduate students are assigned a complementary paper — a rebuttal, a study using a different method, or an earlier or later result on the same question — that the class has not read. Graduate students read it closely and bring it into their small group, helping undergraduates compare what the two papers found and reconcile any differences.

Final presentation– 6850 additional requirements: Graduate students present individually for 20–25 minutes, with 5 minutes for questions. In addition to the undergraduate requirements, they use 2–3 additional papers to provide context on what was understood before their primary paper, what it changed, and/or where it remains contested/uncertain, e.g. preceding papers that the study builds on, competing papers that reach conflicting conclusions or use a different methodology to arrive at the same conclusion, or subsequent papers that test, refute, or extend its conclusions. They will also identify an important unresolved question or direction for future research.

Late and missed assignments: You are expected to attend every class and complete all assignments on-time, but sickness and unavoidable conflicts happen. Please see the italicized text in the assignments section for the corresponding late/missed policy. Valid reasons for absence include personal or family illness, a major religious holiday, university-sanctioned athletic event, scientific conference, or a job interview. Extenuating circumstances that result in multiple missed assignments will be addressed on a case-by-case basis.

Use of Generative AI: Many of us use generative AI to organize and discuss our thoughts, collate information, or proofread writing (including Prof. Miller). However, it is very, very important that we do not get in the habit of relying on AI for quick answers to complex questions in lieu of engaging our own creativity and critical thinking skills. Strengthening these skills will be crucial for building a deep understanding of class material that you'll need for the quizzes, in-class discussions, and presentations, as well as for your future career. Please ensure that all submitted writing is done by you, in your own words. Copy-pasting AI outputs is never acceptable. Do not have AI "read" a paper for you – always read the paper yourself first, and then if you want to, use AI to engage with your understanding of it.

Academic integrity: You are encouraged to discuss concepts and approaches with peers, the instructor, and even AI, but all submitted work must be your own work and include citations where applicable. Please follow the Cornell University Student Code of Conduct and Code of Academic Integrity.

Student Disability Accommodation: Disability accommodations should be submitted through Student Disability Services (https://sds.cornell.edu/). Please note that for accommodations on the quizzes, the accommodation must be submitted at least 5 days in advance of the exam date.

Schedule (Subject to change):

Week 1

Class 1 (Jan 26): Course overview, overview of ocean circulation

Reading and Canvas Quiz 1 assigned: Vallis Chapters 2-3

Class 2 (Jan 28): Dynamics (Coriolis, Ekman, Gyres, western boundary currents)

Week 2

Class 3 (Feb 2): Dynamics continued

Class 4 (Feb 4): Wind-driven ocean circulation

Reading and Canvas Quiz 2 assigned: Vallis Chapter 4-5

Week 3

Class 5 (Feb 9): Ocean density, thermohaline circulation

Class 6 (Feb 11): AMOC

Week 4

(Feb 16): **February Break**

Class 7 (Feb 18): How to read a scientific paper

Reading and Canvas Quiz 3 assigned: Caeser et al (2018)

Week 5

Class 8 (Feb 23): **Quiz 1**

Class 9 (Feb 25): The great AMOC shutdown debate (In-class activity on Caesar et al 2018)

Week 6

Class 10 (Mar 2): The great AMOC shutdown debate continued (North Atlantic Warming Hole papers)

Reading and Canvas Quiz 4 assigned: Baker et al. 2025

Class 11 (Mar 4): The great AMOC shutdown debate continued

Week 7

Class 12(Mar 9): In-class discussion of Baker et al

Class 13 (Mar 11): The great AMOC shutdown debate continued (Bonan et al, 2025, Portman et al, 2026)

Week 8

Class 14 (Mar 16): **Quiz 2**

Class 15 (Mar 18): SOMOC

Reading and Canvas Quiz 5 assigned: Olivier and Haumann 2025

Week 9

Class 16 (Mar 23): The mighty Southern Ocean CO_2 sink

Class 17 (Mar 25): In-class discussion of Olivier and Haumann 2025

Reading and Canvas Quiz 6 assigned: Jacobs et al 2022

Spring break

Week 10

Class 18 (Apr 6): The freshening and slowing of SOMOC

Class 19 (Apr 8): In-class discussion of Jacobs et al 2022

Week 11

Class 20 (Apr 13): AMOC + SOMOC review

Class 21(Apr 15): **Quiz 3**

Final presentation paper choice due; "First-pass" summary due on Canvas

Week 12

Class 22 (Apr 20): Ocean deoxygenation

Class 23 (Apr 22): Ocean deoxygenation continued

Week 13

Class 24 (Apr 27): Student presentations

Class 25 (Apr 29): Student presentations

Week 14

Class 26 (May 4): Student presentations

Class 27 (May 6): Student presentations

Week 15 Last class/Study days begin

Class 28 (May 11): Student presentations/Class wrap-up

Final examination period (May 15 – 22)

Final exam (date and time TBD)